

Uddipan Basu Bir

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PROFESSIONAL SUMMARY

I am in the final phase of my M.Sc. in Data Science, specializing in AI and ML, with nearly five years of experience across Data Analytics, Data Engineering, and Applied AI. At Siemens, Schaeffler, and TCS, I have built end-to-end AI/ML solutions, production-ready RAG and NLP applications, ETL pipelines, dashboards, and automation workflows using Python, SQL, PySpark, Power BI, cloud platforms, and modern AI frameworks. I enjoy solving complex problems and building scalable solutions that run reliably in production.

EXPERIENCE

Munich Music Labs, TUM Germany

Research Team Member

Munich, Germany

Nov 2025 – Present

- Developing a multimodal audio-MIDI model for instrument assignment.

Siemens AG

Working Student – Applied AI Engineer

Erlangen, Germany

May 2025 – Present

- Built an AI reporting pipeline on unified departmental data in a single Snowflake reporting layer, generating automated executive PDF/PPT summaries.
- Developed and deployed Bidify, a RAG-based bid-analysis app, on AWS ECS/ECR with CI/CD, using Texttract, Bedrock, and pgvector.

Tools & Technologies: Snowflake, dbt, AWS AgentCore, Azure ML Studio, Azure OpenAI, LangChain, PyTorch, Azure AI Search, AWS (ECS, ECR), Docker, Langfuse, Postman.

Schaeffler AG

Working Student – Data Analytics and Machine Learning

Nürnberg, Germany

Jun 2024 – Apr 2025

- Built an anomaly detection pipeline to identify unusual BOM component prices and surface flagged cases in Power BI for analysis.
- Developed a quantity forecasting solution using regional market data, growth factors, and historical planning inputs to generate country-level demand scenarios.

Tools & Technologies: Python, Power BI, Power Apps, Power Automate, SAP, Dataverse, Microsoft Power Platform.

Tata Consultancy Services

Data Engineer

Kolkata, India

Sep 2018 – Jun 2022

Sep 2021 – Jun 2022

- Built and maintained 10+ ETL workflows in Ab-Initio using Autosys scheduling, improving data delivery SLA by 30%.
- Optimized ETL processes by migrating logic from Ab-Initio to PySpark, reducing runtime by 20%.

Data Analyst

Sep 2018 – Aug 2021

- Developed Power BI dashboards and reports (MRR, ARR, and P&L), enabling faster data-driven decisions.
- Automated data quality checks and implemented cleaning scripts, reducing manual effort by 20%.

Tools & Technologies: Kafka, Airflow, GitHub Enterprise, Python, Unix, Autosys, Ab-Initio, SQL Indexing, Window Functions, CTEs, SQL Procedures, ServiceNow, Power BI, Power Apps, SQL.

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg

M.Sc. in Data Science; Specialization in Artificial Intelligence & Machine Learning

Erlangen, Germany

Oct 2022 – Present

- **Master Thesis:** Comparative analysis of vision-language models for OCR-based structured JSON extraction from document images.
- **Master Seminar Project (REHAU & FAU):** Forecasting and anomaly detection in time series data for polymer extrusion process. [GitHub](#)

Maulana Abul Kalam Azad University of Technology

Bachelor of Technology in Computer Science & Engineering

Kolkata, India

Jun 2014 – Jul 2018

- **Bachelor Thesis:** Emotion Detection Through Facial Expressions. [GitHub](#)

SKILLS

Programming Languages: Python, SQL, Unix Shell Scripting, TypeScript

AI / GenAI: Hugging Face Transformers, LangChain, LangGraph, Groq Cloud, Azure OpenAI, Azure AI Foundry, AWS Bedrock, SWIFT, LLaMA-Factory, NLTK, spaCy, vLLM, LangFlow, n8n, MCP

Machine Learning & Deep Learning: PyTorch, PyTorch Lightning, Scikit-learn, XGBoost, statsmodels

Vector Databases: pgvector, OpenSearch, Qdrant, FAISS, Chroma

Data Engineering & Orchestration: PySpark, Airflow, Autosys, Ab-Initio

Cloud, ML Platforms & HPC: Azure, AWS, GCP, HPC, SLURM, Azure ML Studio, AWS SageMaker

Experiment Tracking, Evaluation & Observability: Opik, MLflow, Weights & Biases, TensorBoard, Prometheus, Grafana

Backend / FullStack: FastAPI, Flask, Node.js, Streamlit, REST

DevOps & Containerization: Docker, Kubernetes, GitLab CI / GitHub Actions, Jenkins, Minikube, ArgoCD

PROJECTS

Master Thesis: From Pixels to Structure | [GitHub](#)

- Fine-tuning and comparative analysis of multimodal and OCR-based vision-language models using PEFT methods such as LoRA, QLoRA, and Unsloth for structured JSON extraction from handwritten and printed document images.
- Tools & Technologies:** Vision-Language Models, OCR, PEFT, LoRA, QLoRA, Unsloth, Structured JSON Extraction.

Multi-modal, multi-view, multi-task Industrial Quality Inspection | [GitHub](#)

- Developed a multi-modal, multi-view, multi-task deep learning pipeline for classification and quality monitoring in industrial manufacturing.
- Tools & Technologies:** Deep Learning, Multi-Modal Learning, Multi-View Learning, Multi-Task Learning, Computer Vision, Industrial Quality Monitoring.

Self-/Semi-Supervised Industrial Inspection Study | [GitHub](#)

- Built and compared self-supervised, semi-supervised, and combined deep learning approaches for image classification in industrial visual inspection of electrical motor manufacturing.
- Tools & Technologies:** Self-Supervised Learning, Semi-Supervised Learning, Deep Learning, Image Classification, Computer Vision, Industrial Visual Inspection.

Local LLM-based RAG System for Personal Documents | [GitHub](#)

- Built a local, privacy-preserving RAG system for German bureaucracy letters using Streamlit, Ollama, OpenSearch hybrid retrieval, Redis caching, OCR-based ingestion, and Prometheus/Grafana observability, fully deployed via Docker.
- Tools & Technologies:** Streamlit, Ollama, OpenSearch, Hybrid Search, Redis, Prometheus, Grafana, Docker, OCR.

Accident Severity Prediction from Sentiment Embeddings | [GitHub](#)

- Built a sentiment analysis pipeline on news reports to predict accident severity by comparing GloVe, Word2Vec, FastText, and Sentence Transformers embeddings with a Random Forest classifier.
- Tools & Technologies:** Sentiment Analysis, GloVe, Word2Vec, FastText, Sentence Transformers, Random Forest, Text Embeddings.

HACKATHON

Agentic Fraud Detection Hackathon – [Reply](#) | [GitHub](#)

- Built three independent agentic fraud-detection pipelines using LangChain, Groq, and Anthropic SDK for a multi-modal financial dataset spanning transactions, SMS, email, and audio, with dual-agent review patterns and Langfuse observability tracing.
- Designed a cost-optimized three-tier LLM cascade (rule-based screen → Claude Haiku → Claude Sonnet escalation), reducing inference cost to ~5% of a naive approach, with local audio transcription via faster-whisper and parallel batch processing.

LANGUAGES

English: Fluent **German:** Intermediate **Bengali:** Native **Hindi:** Fluent